

Lab 5 NOTES

MATERIALS:

- 1) pH meter & electrode assembly
- 2) magnetic stirrer & stir bar w/ retriever
- 3) 10-mL beaker
- 4) 250 mL beaker
- 5) 250 mL erlenmeyer flask
- 6) Two burets & buret reader
- 7) 25-mL pipette
- 8) wash bottle (dw)
- 9) NaOH soln (known [])
- 10) HCl soln - unknown []
- 11) phenolphthalein

Calibration

pH meter, glass electrode, 10 mL beaker & buffer solns

1) remove connector from pH meter. Clean then pull.

2) Rinse 10 mL beaker w/ 12 mL buffer → add 5.0 mL buffer to beaker & record pH.

3) Lower cleaned electrode into beaker & swirl beaker to replace dw film on glass bulb.

• determine room temp from barometer & turn temperature till matches barometer. - Now it is calibrated

- done at time of pH meter

Store electrode in distilled water beaker when not using

Buffer 1:

our electrode
stabilized at

7.01 & 10.02

Titration: HCl + NaOH, no pH meter

1) 250 mL beaker w/ 200 mL NaOH

record []

$[NaOH] = 0.22 M$

0.2200

2) remove water from buret

- can rinse w/ NaOH.

3) put NaOH beaker under Buret & let excess NaOH into the beaker w/ air bubble under stopcock & gone.

Buret should reach 0.00 mL mark.

- wait 10 seconds → remove beaker & record initial

buret reading

~ 150 mL

4) Place 250 mL beaker w/ HCl soln and stirrer or stirrer

5) Add HCl to 2nd Buret

6) w/ pipette measure 25 mL of HCl to clean 250 mL

beaker. Then add 75 mL of dw, 34 drops - phenolphthalein & stir bar & place on stirrer.

- increase speed till small vortex appears.

- record initial buret reading & HCl soln color

buret	0.00	23.15	23.15
color	clear	clear	23.15

7) start adding NaOH to HCl w/ buret.

equiv pt: around slightly pink color - slow down & start adding

NaOH from other buret by the drop

nequr pH / pink color should stay for 30 sec.

then turn off stirrer & stop

do this 2x

initial NaOH buret reading: 0.0, clear

Final NaOH buret reading: 23.80, pink
error = 2x

Titration #2: HCl against NaOH w/ pH meter
- can use phenolphthalein if want

- refill & release NaOH $\bar{E}$ buret - no air bubbles

1) clean 250 mL beaker add 25 mL of HCl, 75 mL
dw & 3-4 drops phenolphthalein to beaker.
turn on stirrer

2) rinse glass electrode w/ dw & dry w/ Kimwipe.
- position electrode away from stirrer.

3) record initial pH, color, & initial buret reading to
nearest 0.02 mL

4) start titration add NaOH till pH change by ~ 0.3
plot vol of NaOH vs pH.
record & titrate every 0.3 units till S.S pH or
higher.

5) near equiv point add NaOH dropwise & record pH.
after endpoint, keep adding ~~at~~ by 0.5 till pH=11
Then add more NaOH till pH=11.5 ± 0.2

Record to nearest 0.02 mL

Any
read
when
pH change
by 0.3

No pH meter Titration:

Run	Init Buret Reading ^(mL)	Final Buret Reading ^(mL)	ΔV mL
1			
2			
3			
4			
Avg			

by pH meter:	Vbase ^(mL) initial	pH	ΔV (mL)	ΔpH	$\Delta pH / \Delta V$ (1/mL)
(MIT)	0.0	2.00	—	—	—
	16.05	2.32			
	20.20	2.61			
	21.72	2.90			
	22.40	3.22			
	22.65	3.57			
	22.80	3.95			
	22.85	4.24			
	22.88	5.03			
	22.90	5.30			
	22.92	5.70			
	22.98	6.05			
	23.00	6.48			
	23.04	6.86			
	23.10	8.93			
	23.12	9.34			
	23.18	9.64			
	23.24	9.87			
	23.32	10.11			
	23.50	10.44			
	23.75	10.72			
	24.25	10.98			
	25.56	11.31			

equiv reached
pink

clean